A Practitioner's Reality Check: Classical CFD vs. Neural Surrogates on a Single Consumer GPU

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***Abstract*—Recent large-scale benchmarking (e.g., REALM [1]) shows neural surrogates for multiphysics flows often report strong nominal accuracy while missing key transient structures. This paper asks a narrower, practitioner-oriented question: on the kind of small, well-posed problem a CFD engineer encounters daily, how does a classical linear reduced-order model (POD-Galerkin, DMD) compare to a modern neural surrogate (NVIDIA PhysicsNeMo), when both run end-to-end on a single consumer workstation? We report three self-contained experiments — parametric interpolation via POD-ROM vs. a fully-connected neural surrogate, transient analysis via DMD, and interface prediction via a CNN on a two-phase (VOF) flow — all on a single RTX 4090 GPU. Under conditions favorable to linear ROM, classical methods remain substantially more accurate and equally fast; neural methods degrade further under autoregressive rollout on sharp-interface data. We state these limitations explicitly and position this as an early, practitioner-level data point rather than a comprehensive benchmark, drawing on nine years of industrial CFD experience (Siemens STAR-CD/STAR-CCM+).**

***Index Terms*—CFD, neural surrogates, reduced-order modeling, proper orthogonal decomposition, dynamic mode decomposition, benchmarking, HPC on consumer hardware**

I. Introduction

Neural surrogate modeling for computational fluid dynamics has advanced rapidly, but recent large-scale evaluations suggest the field may be suffering from an "illusion of mastery": existing evaluations rely too heavily on simplified, low-dimensional proxy tasks that fail to expose the fragility of neural models under realistic conditions. The REALM benchmark [1] operationalized this concern at scale, testing over a dozen neural surrogate families across eleven high-fidelity multiphysics datasets, and found a persistent gap between nominal accuracy metrics and physically trustworthy behavior.

This paper asks a smaller, more mundane question from the opposite end of the resource spectrum: does the same qualitative gap already show up when a single practitioner, using a single consumer GPU and textbook-scale test cases, runs classical and neural methods side by side? This is not an attempt to replicate REALM's scale. It is closer to a sanity check that any practitioner can run in an afternoon, and a first step toward the kind of HPC-adjacent, resource-constrained validation that this workshop's community engages with regularly.

The motivation comes from nine years of applied CFD work (Siemens, STAR-CD/STAR-CCM+), where the operative question was rarely "which method is more accurate in principle" but "which method can I trust enough to act on today." That practical framing is the lens applied throughout.

A. Related Work

Projection-based reduced-order models such as POD-Galerkin [2] and DMD [3] have a decades-long track record in fluid dynamics, valued for their interpretability and low computational cost, but they typically assume a fixed computational domain and struggle to generalize across geometry [2]. Physics-informed neural networks [4] and neural operators such as the Fourier Neural Operator [5] were introduced in part to address this limitation, offering mesh-agnostic generalization at the cost of interpretability and, as this paper's Experiment C illustrates, robustness under autoregressive rollout on sharp-interface data. A growing body of work has coupled POD with neural networks (POD-NN) to combine the interpretability of linear modes with the flexibility of learned mappings, though these hybrid approaches inherit POD's difficulty with geometrically varying domains unless paired with mesh-morphing or point-cloud-based encoders.

At the benchmarking level, REALM [1] is the closest point of comparison: it systematically evaluates over a dozen neural surrogate architectures on eleven high-fidelity, application-driven multiphysics datasets requiring hundreds to thousands of HPC-hours to generate, and identifies a persistent gap between nominal accuracy and physically trustworthy rollout behavior. Other recent benchmarking efforts in this space similarly emphasize dataset scale and architectural diversity. This paper occupies a complementary and much smaller niche: rather than adding another large-scale benchmark, it asks whether the same qualitative warning is visible to a single practitioner working with consumer-grade hardware and off-the-shelf, open-source tools (OpenFOAM, PhysicsNeMo) in the course of a single afternoon. To our knowledge, no prior work has explicitly framed this resource-constrained, practitioner-level replication as a check on whether large-scale benchmarking findings are artifacts of scale or genuine structural properties of naive supervised training on physical fields.

II. Experimental Setup

A. Hardware & Software

Table I and Table II summarize the workstation used for all experiments. No cluster or commercial license was required; the entire pipeline runs on commodity hardware.

TABLE I
WORKSTATION HARDWARE SPECIFICATION.

Component	Detail
CPU	Intel Core i9-13900F (24c/32t)
RAM	64 GB DDR5-5600
GPU	NVIDIA RTX 4090, 24 GB VRAM
Storage	2× Samsung 990 PRO 1TB NVMe

TABLE II
SOFTWARE STACK.

Layer	Version
WSL2	Ubuntu 26.04 LTS
CFD solver	OpenFOAM v2512 [6]
AI framework	NVIDIA PhysicsNeMo 1.3.0 [7]
Deep learning	PyTorch 2.13.0+cu126

B. Validation

OpenFOAM was validated on the lid-driven cavity (icoFoam) and dam-break (interFoam, VOF) tutorials. PhysicsNeMo's official test suite passed in full (2511/2511, 163 skipped, 0 failed). A KNN benchmark showed cuML scaling far better than naive torch/scipy implementations (6.27 ms vs. 79.9 ms at 100k points/5k queries), relevant to mesh/point-cloud-based surrogates such as DoMINO (Fig. 1). These are installation/sanity checks, not benchmark claims in their own right.

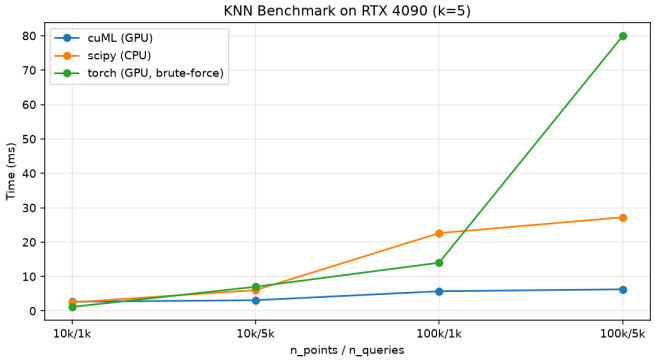


Fig. 1. KNN search time across implementations, RTX 4090, $k = 5$.

III. Experiment A — POD-ROM vs. Neural Surrogate

Setup. Lid-driven cavity, parametrized by lid velocity $U \in \{1, \ldots, 10\} \setminus \{7\}$ (9 OpenFOAM runs training, held-out $U = 2.5, 7.5$).

POD-ROM formulation. Given snapshots $\mathbf{u}(x, y; U_i) \in \mathbb{R}^{1200}$ for $i = 1, \ldots, 9$, we assemble the snapshot matrix $\mathbf{S} = [\mathbf{u}_1, \ldots, \mathbf{u}_9] \in \mathbb{R}^{1200\times 9}$ and compute its singular value decomposition $\mathbf{S} = \mathbf{\Phi\Sigma V}^\top$. The velocity field is approximated as

$$\hat{\mathbf{u}}(x, y; U) \approx \sum_{k=1}^{r} a_k(U)\, \boldsymbol{\phi}_k(x, y), \tag{1}$$

where $\boldsymbol{\phi}_k$ are the leading $r = 3$ left singular vectors (POD modes) and the modal coefficients $a_k(U)$ are obtained by projecting each training snapshot onto the mode basis and interpolating across U with natural cubic splines. This is a fully data-driven (non-intrusive) ROM: no Galerkin projection of the Navier–Stokes operator is performed, keeping the comparison to the neural surrogate on equal footing in terms of required domain-specific derivation.

Neural surrogate. PhysicsNeMo FullyConnected network $f_\theta : (x, y, U) \mapsto (u, v)$ with 4 hidden layers of 128 units each and $\tanh$-family activations, trained for 3000 epochs on 3600 collocation points (400 per training case × 9 cases) with MSE loss, Adam optimizer, and cosine learning-rate decay from an initial rate of 10^{-3}. Final training loss was 5.90×10^{-4}. Unlike the POD-ROM, the network receives no explicit modal structure and must learn the mapping from raw coordinates and the parameter U directly.

TABLE III
POD-ROM VS. NEURAL SURROGATE.

Case	POD err.	POD time	NN err.	NN time
$U = 2.5$	0.128%	0.056 ms	3.364%	0.276 ms
$U = 7.5$	0.018%	0.034 ms	2.199%	0.264 ms

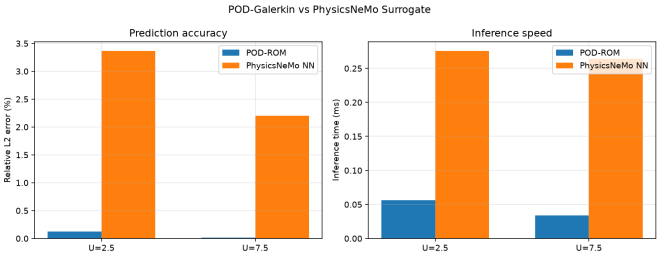


Fig. 2. Accuracy and inference-speed comparison, POD-ROM vs. PhysicsNeMo.

POD-ROM outperformed the neural surrogate by roughly 25× in accuracy at comparable or better speed. **This should not be read as "classical beats neural."** The problem was deliberately smooth, single-geometry, single-parameter, low-Reynolds — precisely the regime a linear ROM dominates (dominant POD mode $R^2 = 0.99999$ linear in U). Neural methods are expected to win when geometry varies, the parameter space is high-dimensional, gradients are needed for optimization, or physics is strongly nonlinear — none of which apply here. With 9 training snapshots and 2 test points, this is a single reproducible data point, not a statistical claim.

IV. Experiment B — Dynamic Mode Decomposition

DMD formulation. Given a sequence of $m = 100$ time-resolved snapshots $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ sampled at fixed $U = 5$ with $dt = 0.005\,\mathrm{s}$, we form the data matrices $\mathbf{X} = [\mathbf{u}_1, \ldots, \mathbf{u}_{m-1}]$ and $\mathbf{X}' = [\mathbf{u}_2, \ldots, \mathbf{u}_m]$ and seek the best-fit linear operator $\mathbf{A}$ satisfying $\mathbf{X}' \approx \mathbf{A}\mathbf{X}$. Exact DMD [3] computes the eigendecomposition of $\mathbf{A}$ restricted to the r-dimensional POD subspace of $\mathbf{X}$, yielding DMD modes $\boldsymbol{\varphi}_j$, eigenvalues λ_j, and continuous-time growth rates/frequencies via $\omega_j = \ln(\lambda_j)/dt$. We truncate to $r = 5$ modes, retaining 99.999989% of the singular value energy.

Setup and results. Fixed $U = 5$, transient icoFoam run, 100 snapshots. All five retained modes had $\mathrm{Im}(\omega_j) = 0$ (purely decaying, non-oscillatory), consistent with the expected monotonic relaxation to steady state at this low Reynolds number, with no periodic vortex shedding. Mode 1 ($|\lambda| \approx 1$, near-zero growth rate) is the steady-state attractor and reproduces the same primary-vortex pattern seen in the PINN solution of Section III's validation step; mode 0 has the fastest decay rate and is spatially localized near the moving-lid corner, consistent with it capturing the impulsive start transient; modes 2 and 3 have intermediate decay rates and show swirling spatial patterns consistent with transient vortex reorganization before the flow settles. complementary rather than competing [2], [3]: POD-ROM answers "how does the solution change across parameters," DMD answers "how does a single solution evolve in time." Both are cheap and interpretable — properties that matter when an engineer must explain *why* a result looks the way it does.

TABLE IV
DMD mode decomposition of the transient cavity flow.

Mode	$\lvert\lambda\rvert$ amp.	Growth (1/s)	Freq. (Hz)	Interpretation
1	21.48	-0.0014	0.0	Steady-state attractor
0	11.05	-461.79	0.0	Fastest-decaying shock
2	25.40	-122.08	0.0	Vortex reorganization
3	24.67	-98.52	0.0	Vortex reorganization
4	8.98	-44.33	0.0	Slowest-decaying

V. Experiment C — Neural Interface Prediction

Dam-break (interFoam), 97 snapshots of `alpha.water` [8] on a 64×64 grid. 5-layer CNN encoder-decoder, 2000 epochs.

Architecture and training. The volume-fraction field $\alpha(x, y, t) \in [0, 1]$, tracking the liquid-gas interface via the VOF method [8], was interpolated from the unstructured OpenFOAM mesh onto a regular 64×64 grid at each of 97 time steps. A 5-layer convolutional encoder-decoder $g_\theta : \alpha_t \mapsto \hat{\alpha}_{t+1}$ (Conv2d layers with stride-2 downsampling in the encoder, transposed-convolution upsampling in the decoder, sigmoid output activation to respect $\alpha \in [0, 1]$) was trained on 81 consecutive frame pairs $(t, t + 1)$ with 15 held out for testing, using MSE loss and 2000 training epochs (final training loss 1.29×10^{-3}). Two evaluation protocols were used: (i) *single-step*, where the model receives the true previous frame at every test time, isolating one-step prediction error; and (ii) *autoregressive rollout*, where the model receives only the initial seed frame and is then repeatedly fed its own predictions for 15 consecutive steps, the regime relevant to actually replacing rather than merely post-processing a solver.

TABLE V
NN-VOF: single-step vs. rollout.

Mode	Rel. L2 error	Speed
Single-step	13.5%	0.24 ms/step
Rollout (15 steps)	44.6%	0.24 ms/step

Single-step error (13.5%) was already $\sim 5\times$ worse than Experiment A's neural surrogate (2–3%), compounding to 44.6% under rollout, with visible blurring and loss of fine droplet structure (Fig. 3). This is the well-known failure mode of MSE-trained image-to-image regression on sharp-interface data, reproducing at toy scale the motivation for specialized approaches (phase-field PINNs [4], level-set neural operators [5]) rather than arguing against them.

VI. Discussion

A. Scope of the claim

This does **not** show neural surrogates are inferior in general — the problems were chosen to be exactly the regime classical ROM is built for. It **does** show the accuracy gap can be large ($\approx 25\times$) on identical hardware in comparable wall-clock time, invisible from inference speed alone. The operational takeaway for practitioners: match the tool to the geometry/parameter regime rather than assuming a speed advantage implies an accuracy advantage.

B. Threats to validity

Textbook-scale cases only; no industrially complex geometry. Small samples (9 snapshots, single Reynolds number). No hyperparameter search on the neural models. Single-seed runs.

C. Toward a geometry-varying extension

The single largest limitation of the POD-ROM formulation used here is implicit in Eq. (1): it assumes every snapshot lives on the same discretization, so the modes $\boldsymbol{\phi}_k$ are only meaningful when the computational domain is fixed. This is a well-documented limitation of projection-based ROMs [2] and is precisely the condition this paper's test cases were chosen to satisfy, in order to isolate the accuracy comparison from confounds introduced by mesh morphing or point-cloud encoding. A natural next step — left for future work — is to introduce a parametrized geometric obstacle into the cavity or channel flow and measure the deformation magnitude at which a mesh-agnostic neural surrogate (e.g., a point-cloud or graph-based architecture such as PhysicsNeMo's DoMINO) begins to outperform a POD-ROM equipped with a mesh-morphing preprocessing step. Identifying this crossover threshold, rather than simply confirming that it exists, would convert the qualitative expectation stated in Section III into an actionable,

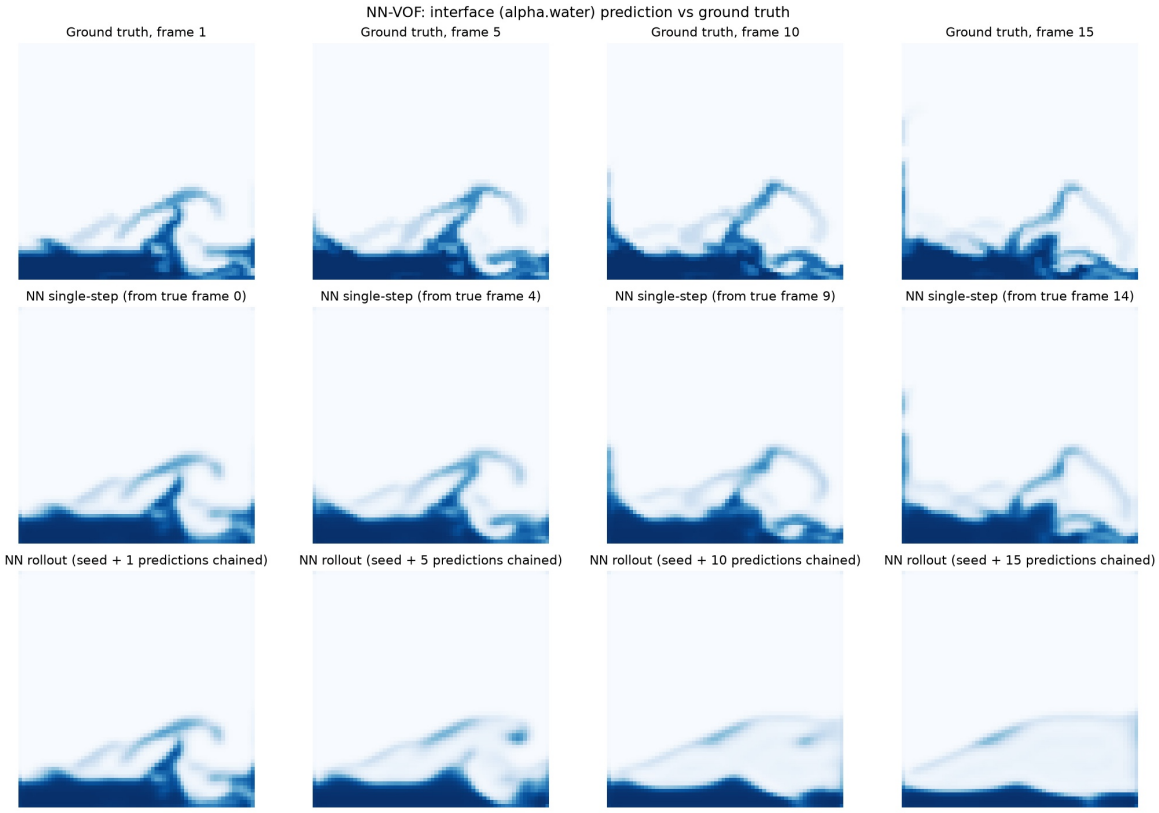


Fig. 3. NN-VOF prediction vs. ground truth. Top: ground truth. Middle: single-step. Bottom: autoregressive rollout. Progressive blurring is visible under rollout, consistent with Table V.

quantitative design rule for practitioners choosing between ROM and neural approaches under geometric variability.

D. Implications for HPC-adjacent scientific ML practice

This community routinely evaluates AI-for-science methods at extreme scale — across large clusters, diverse accelerators, and high-fidelity datasets requiring substantial HPC allocation. That scale is essential for characterizing state-of-the-art surrogate performance, but it also raises the cost of even a first sanity check: a practitioner deciding whether to invest a research group's compute budget in a neural surrogate for a new problem class may not have HPC-scale resources available at the outset. The experiments here suggest that a meaningful first signal — whether a naive neural surrogate is even competitive with a decades-old linear baseline on a simplified version of the target problem — can be obtained on commodity hardware before committing to larger-scale validation. We do not propose this as a substitute for rigorous, large-scale benchmarking such as REALM, but as a complementary, low-cost triage step: if a neural surrogate cannot outperform a linear ROM on a simplified, favorable-case version of a problem, that is useful information before scaling up. Making this kind of check cheap and habitual may also help surface the REALM-style accuracy/trustworthiness gap earlier in a project's lifecycle, when it is less costly to change course. In our experience managing CFD workloads in an industrial setting, the cost of discovering an accuracy/trustworthiness gap late — after a surrogate has already been integrated into a design-optimization loop or a real-time monitoring pipeline — is substantially higher than the cost of a consumer-GPU sanity check performed up front, even accounting for the sanity check's inherent limitations in scale and statistical power.

E. Relation to prior work

REALM [1] identified, at scale, a persistent gap between nominal accuracy and physically trustworthy behavior. Experiment C reproduces the same qualitative pattern with a single CNN, a single GPU, and an afternoon of compute — evidence, though not proof, that the gap REALM documents is not purely an artifact of scale. A single architecture is not conclusive; broader validation across architectures and problem classes, closer to REALM's own scope, would be needed to establish this reading rigorously. We view this note as a cheap, easily repeated observation consistent with REALM's finding, offered to this workshop's community as an entry point for discussion of low-cost, HPC-adjacent validation practices for scientific ML.

VII. Conclusion

Tool selection should match problem structure, not hype. On this class of problem, classical linear methods (POD-ROM, DMD) remain highly competitive on accuracy and

speed, and their interpretability is an operational advantage where engineers must justify results. Neural surrogates' value lies elsewhere — geometry generalization, high-dimensional parameter spaces, differentiability for optimization — not raw accuracy on every problem. The entire pipeline runs on a single consumer GPU workstation with open-source tools, suggesting this kind of practitioner-level sanity check is broadly replicable without institutional compute access, and may be a useful complement to large-scale HPC benchmarking efforts within this community.

Data Availability

Code for the POD-ROM, DMD, and CNN experiments will be made publicly available upon acceptance.

Author Note

The author spent nine years as an industrial CFD engineer (Siemens, STAR-CD/STAR-CCM+) prior to transitioning to HCI and perceptual-science research, which motivates the practitioner framing of this note.

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